Battery Care and Safety

Know your battery

Northwind e-bikes use two lithium-ion packs. Both lock into the down tube with the same key as the frame lock.

- **NW-BAT-500**: 36 V, 13.9 Ah, 500 Wh. Standard on the Larkspur commuter. Weighs 6.4 lb (2.9 kg). Typical range 30-55 miles.
- **NW-BAT-720**: 48 V, 15 Ah, 720 Wh. Standard on the Ridgeline and Haul 2. Weighs 8.8 lb (4.0 kg). Typical range 40-70 miles.

The two packs are **not interchangeable**. A NW-BAT-720 won't seat in a Larkspur frame, and the controllers are voltage-specific.

Both batteries charge with the **NW-CHG-4A** charger (4 amp output). A full charge from empty takes about 4.5 hours for the NW-BAT-500 and 6 hours for the NW-BAT-720. A spare charger costs $79.

Charging

- Charge only between **32 °F and 104 °F (0-40 °C)**. Charging a cold battery causes permanent damage to the cells. If the bike has been outside in winter, bring the battery indoors and let it warm up for at least 2 hours before plugging in.
- Charge on a hard, non-flammable surface, away from beds, sofas and curtains.
- Don't charge while you're asleep or away from home. Unplug when the charger LED turns green.
- Plug the charger into the wall first, then into the battery. Unplug in the reverse order.
- Never use a charger that isn't a genuine NW-CHG-4A, even if the connector fits.

You can ride in temperatures from 14 °F to 113 °F (-10 °C to 45 °C). Expect up to 30% less range below 40 °F.

Charger LED codes

The LED on the NW-CHG-4A tells you what's happening:

LED	Meaning	What to do
Solid red	Charging	Nothing, normal
Solid green	Fully charged, or no battery connected	Unplug
Fast-flashing red	Battery temperature out of range	Move to a room at 50-77 °F and retry after 2 hours
Slow-flashing amber	Cell imbalance or communication fault	Unplug for 10 minutes, reseat the battery, retry once; if it repeats, stop and contact support

LED	Meaning	What to do
Alternating red and green	Charger fault	Stop using the charger and request a replacement
No light	No mains power	Check the wall outlet and the AC cable

Storage

If you won't ride for more than two weeks:

1. Charge or discharge the battery to **40-60%** (two to three bars on the battery's own button gauge).
2. Remove it from the bike and store it indoors at 50-77 °F (10-25 °C).
3. Check it every **3 months** and top it back up to 40-60%.

Storing a battery at 100% or fully empty for long periods accelerates aging. A pack left at 0% for several months can enter deep-discharge protection and may not wake up again.

Lifespan

Both packs are rated for **800 full charge cycles** before capacity falls to about 70% of new. A full cycle is the equivalent of charging from 0 to 100%, so two charges from 50% to 100% count as one cycle. Most riders reach 800 cycles in 3-5 years. Partial charges are fine and don't harm the battery.

Cleaning and water

Wipe the battery with a damp cloth. Never use a pressure washer or a hose on the contacts. If the pack has been dropped from more than 3 feet, or the case is cracked, swollen or smells sweet or burnt, stop using it immediately.

Safety notice NW-SN-0412

We've identified a manufacturing issue in a limited batch of **NW-BAT-720** packs produced in November 2023. The cell holder may crack and, in rare cases, cause overheating while charging.

- **Affected serials:** B720-2311-0001 through B720-2311-4800. The serial is on the label under the battery.
- **Not affected:** any NW-BAT-500, and NW-BAT-720 packs with other serial ranges.
- **What to do:** stop charging the battery now. Register at northwindcycles.example.com/safety with your battery serial. We'll ship a free replacement pack and a prepaid hazmat return box within 10 business days. Don't send the battery back in any other packaging.

Affected owners also receive a $50 store credit. This notice doesn't change your warranty dates.

Disposal

Don't put a lithium battery in household trash or recycling. Bring end-of-life packs to any Northwind showroom for free recycling, or request a prepaid recycling box by email at safety@northwindcycles.example.com.